

PRACTICAL 4

CONDITIONS AND BRANCHING

Introduction

This practical covers conditions, branching statements, for loops, while loops, logical operators, and basic number checking in Python.

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Practical	4

Q1. Check if Lionel Messi has more than 10 achievements.

```
players = [
    ["Serena Williams", "Tennis", 23],
    ["Lionel Messi", "Soccer", 7],
    ["Michael Phelps", "Swimming", 23],
    ["Usain Bolt", "Athletics", 8],
    ["Roger Federer", "Tennis", 20],
    ["Cristiano Ronaldo", "Soccer", 5]
]

for player in players:
    if player[0] == "Lionel Messi":
        if player[2] > 10:
            print("Player Name:", player[0])
            print("Sport:", player[1])
            print("Achievements:", player[2])
        else:
            print("Lionel Messi does not have more than 10 achievements.")
```

Output:

Lionel Messi does not have more than 10 achievements.

Q2. Check if a player belongs to Tennis or has exactly 20 achievements.

```
for player in players:
    if player[1] == "Tennis" or player[2] == 20:
        print("Condition is true for:", player[0])
```

Output:

Condition is true for: Serena Williams
Condition is true for: Roger Federer

Q3. Check if a player has less than 10 achievements and does not play Tennis.

```
for player in players:
    if player[2] < 10 and player[1] != "Tennis":
        print("Player Name:", player[0])
        print("Sport:", player[1])
        print("Achievements:", player[2])
```

Output:

Player Name: Lionel Messi
Sport: Soccer
Achievements: 7

Player Name: Usain Bolt
Sport: Athletics
Achievements: 8

Player Name: Cristiano Ronaldo
Sport: Soccer
Achievements: 5

Q4. Take an integer input (1 to 9999) and determine even/odd, palindrome, and divisibility by the sum of its digits.

```
num = int(input("Enter a number between 1 and 9999: "))

# Check even or odd
if num % 2 == 0:
    print("The number is even.")
else:
    print("The number is odd.")

# Check palindrome
if str(num) == str(num)[::-1]:
    print("The number is a palindrome.")
else:
    print("The number is not a palindrome.")
```

```

# Find sum of digits
temp = num
sum_digits = 0

while temp > 0:
    digit = temp % 10
    sum_digits = sum_digits + digit
    temp = temp // 10

print("Sum of digits:", sum_digits)

# Check divisibility by sum of digits
if num % sum_digits == 0:
    print("The number is divisible by the sum of its digits.")
else:
    print("The number is not divisible by the sum of its digits.")

```

Output:

```

Enter a number between 1 and 9999: 121
The number is odd.
The number is a palindrome.
Sum of digits: 4
The number is not divisible by the sum of its digits.

```

Q5. Print all elements between -5 and 5 using the range() function.

```

for i in range(-5, 6):
    print(i)

```

Output:

```

-5
-4
-3
-2
-1
0
1
2
3
4
5

```

Q6. Print the given list using a for loop.

```

squares = ['red', 'yellow', 'green', 'purple', 'blue']

for colour in squares:
    print(colour)

```

Output:

```

red
yellow
green
purple
blue

```

Q7. Copy the strings to new_squares and stop the loop if the value is not 'orange'.

```

squares = ['red', 'yellow', 'green', 'purple', 'blue']

new_squares = []

for value in squares:
    if value == 'orange':
        new_squares.append(value)
    else:
        break

print(new_squares)

```

Output:

```
[]
```

Note: The first value is 'red', so the loop stops immediately.

Q8. Using a while loop, find animal names made of 7 letters and create a separate list.

```
Animals = ["lion", "giraffe", "gorilla", "parrots", "crocodile", "deer", "swan"]

new_animals = []

i = 0

while i < len(Animals):
    if len(Animals[i]) == 7:
        new_animals.append(Animals[i])
    i = i + 1

print(new_animals)
```

Output:

```
['giraffe', 'gorilla', 'parrots']
```

End of Practical 4